

JAYANTH DASAMANTHARAO

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PROFESSIONAL EXPERIENCE

AI Engineer - Fedway Associates, EliteUS Software Solutions Aug 2024 - Present | Edison, NJ

- Designed and deployed production GenAI systems powering real customer workflows, leveraging LLMs, embeddings, retrieval-augmented generation to improve task accuracy, relevance, user productivity at scale.
- Developed and iterated on LLM quality using fine-tuning, prompt optimization, and evaluation (BLEU, ROUGE, precision/recall, A/B testing), enabling experimentation and improvements in model performance.
- Built scalable data and retrieval pipelines including document chunking, hybrid semantic search, reranking, and vector-based memory architectures, indexing 10,000+ pages and serving 12,500+ users with low latency.
- Implemented reusable backend ML systems and APIs using Python, FastAPI, and cloud infrastructure, with robust logging, monitoring, authentication, guardrails to support multiple GenAI products in production.

Data Science Intern, Harvest Software Solutions June 2023 - Sept 2023 | Jacksonville, Florida

- Engineered scalable ML pipelines and monitoring systems for large advertising datasets, increasing campaign visibility from 15% to 45% while ensuring reliability and performance.
- Built ML services and NLP APIs using XGBoost, H2O AutoML, and FastAPI, improving sentiment classification accuracy by 65% and campaign spend efficiency by 50%.
- Designed SQL/Python data processing and evaluation pipelines for feature engineering, model training, and validation, enabling rapid experimentation and improving engagement quality by 40%.

Application Development Associate, Accenture June 2021 - Aug 2022 | Hyderabad, India

- Developed and optimized Python-based ETL and data processing pipelines using hash maps, sets, indexed lookups to efficiently clean, deduplicate, and transform large datasets, reducing production issues by 50%.
- Implemented SQL workflows and joins with partitioning, indexing, and window functions, enabling efficient aggregation and analysis over millions of records in AWS Athena and Redshift.
- Applied batching, caching, and algorithmic optimizations for data validation and pipeline execution, improving system performance and reliability by 25%.

TECHNICAL SKILLS

Programming:	Python, R, SQL. Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, FastAPI.
Machine Learning :	Model Fine-tuning, evaluation and experimentation, deployment, Feature Engineering.
GenAI & LLMs:	LLM fine-tuning, RAG, embeddings, prompt optimization, LangChain, multimodal modeling.
AI Platforms/Agents:	Agentic systems, GenAI systems, memory architectures, tool calling, guardrails.
APIs & Services:	REST APIs, API versioning, authentication, developer tooling.
Cloud & MLOps:	PostgreSQL, Data Warehousing, AWS (Bedrock, Athena, Redshift), GCP, Azure, Vectordb.

EDUCATION

Rutgers University, Master of Science in Data Science Sept 2022 - May 2024 | New Brunswick, NJ
Personal Website: <https://jayanthdasamantharao.github.io/>

Andhra University B.Tech in Electrical and Electronics Engineering 2017 - 2021 | Visakhapatnam, India

RESEARCH PROJECTS

Self-Driving Cars (Computer Vision): Built autonomous driving cars with vision pipeline using CNNs, YOLO, Faster R-CNN, achieving 95%+ object detection accuracy.

Multimodal GenAI System (Production): Built LLM-powered retrieval and generation pipelines using embeddings, caching, multimodal inputs, focusing on response quality, robustness, and production reliability.

Bayesian Classification: Implemented Bayesian logistic regression in PyMC3 to predict diabetes status, incorporating informative priors, achieving 0.85 F1-score.

For additional projects and code samples: <https://github.com/JayanthDasamantharao>